



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI

Dataset at a Glance

3,900 Rows

Transactional records

18 Columns

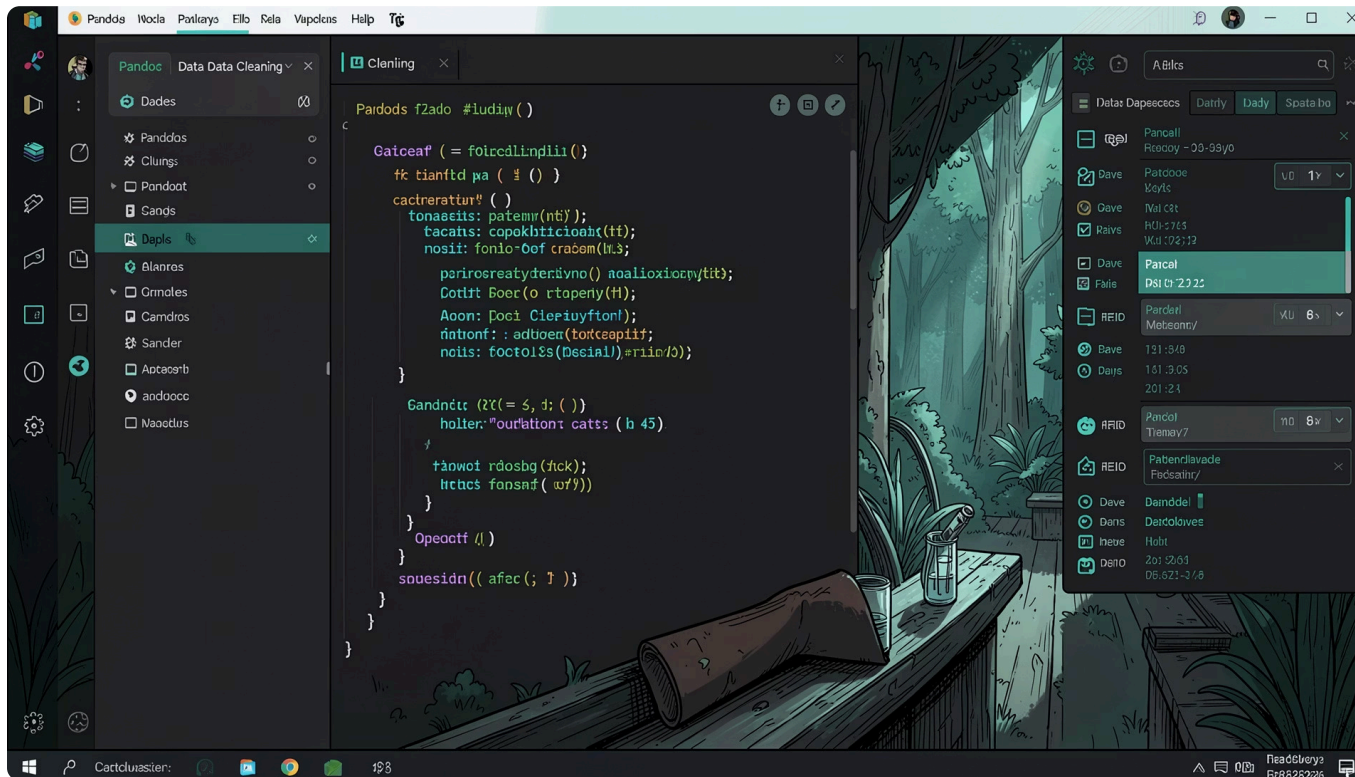
Demographics, purchases & behavior

37 Missing

Review Rating values imputed



Python: Data Preparation



→ Clean & Standardize

Snake_case renaming, median imputation

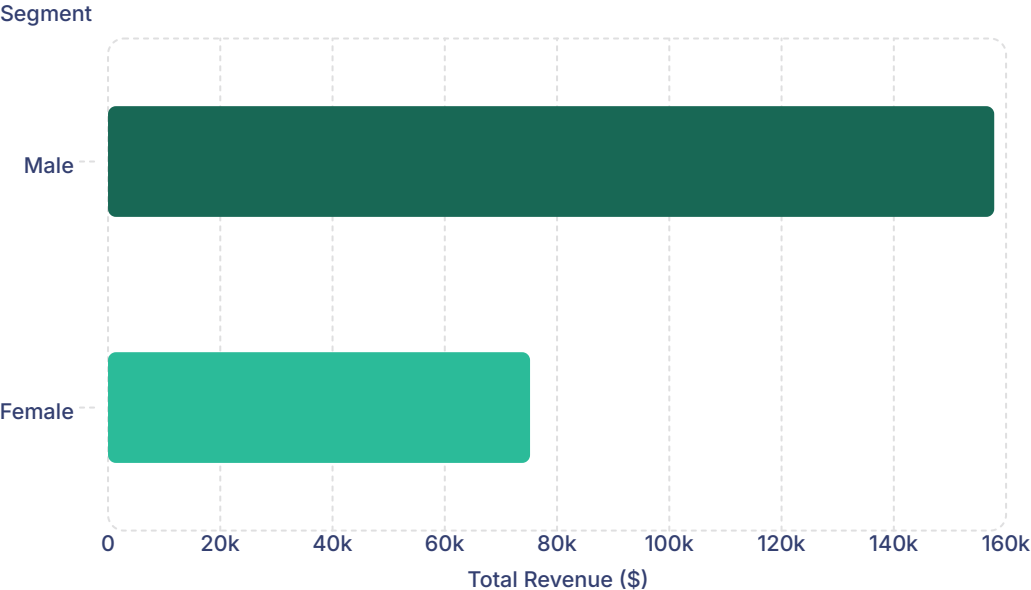
→ Feature Engineering

age_group bins,
purchase_frequency_days

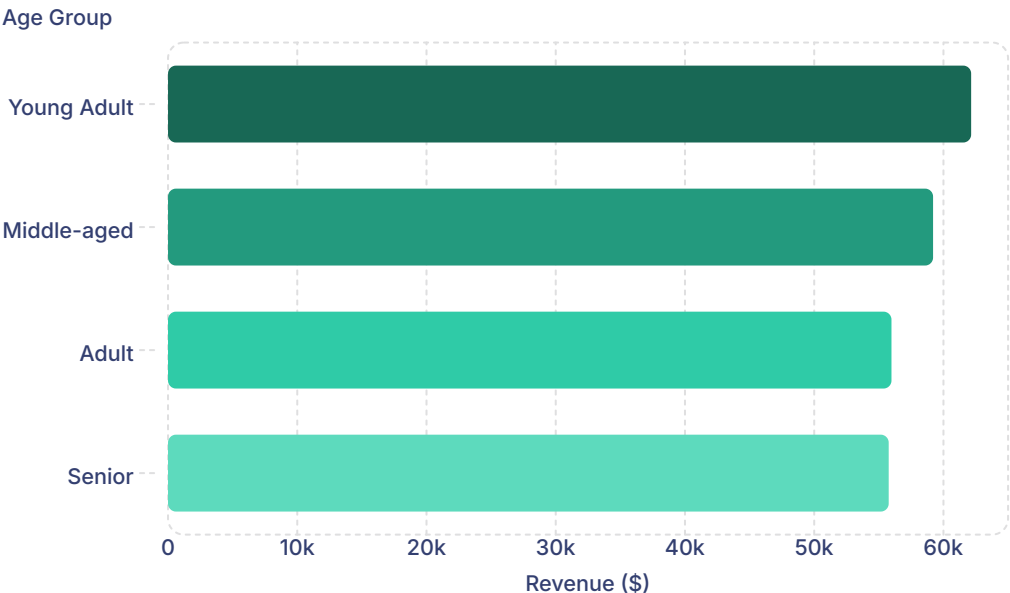
→ Load to PostgreSQL

Cleaned DataFrame ready for SQL

Revenue: Gender & Age

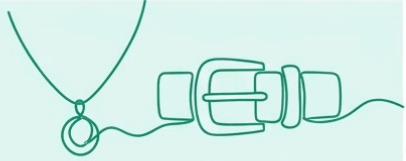


Revenue by Age Group



Male revenue is **2.1× higher**. Age group revenue is evenly distributed.

Top Products & Categories



Accessories:
Jewelry,
Sunglasses,
Belt



Clothing:
Blouse,
Pants,
Shirt



Footwear:
Sandals,
Shoes,
Sneakers



Outerwear:
Jacket,
Coat

Highest Rated

Gloves

3.86 ★

Sandals

3.84 ★

Boots

3.82 ★

Shipping & Discount Insights

\$58.46

Standard Shipping

\$60.48

Express Shipping

50%

Hat Discount Rate

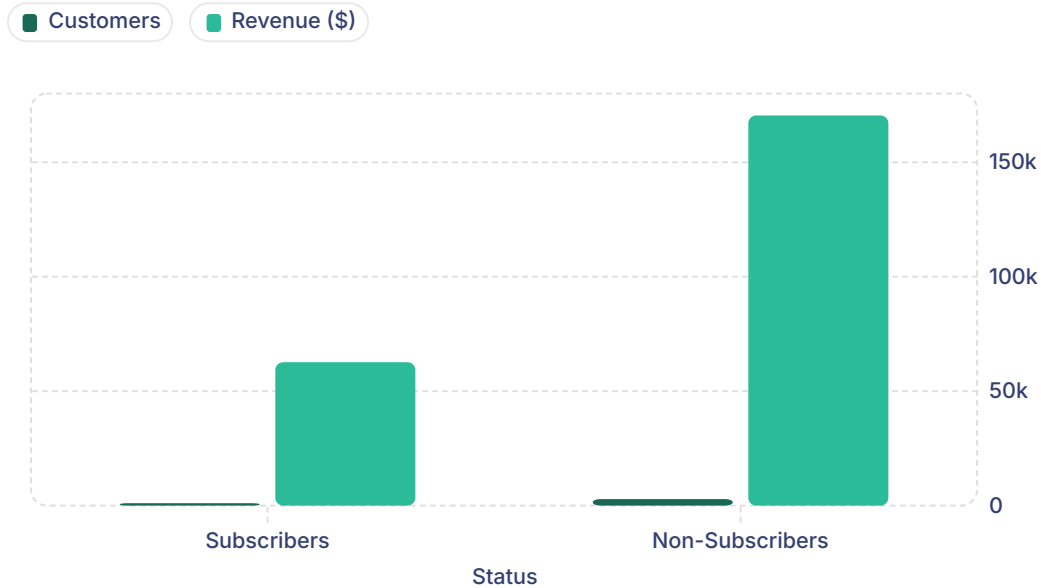
Highest among all products

839

High-Spend Discount Users

Above-average spenders using discounts

Subscribers vs. Non-Subscribers



Key Findings

- Subscribers: **27%** of customers
- Non-subscribers drive **73%** of revenue
- Avg spend nearly equal (~\$59.50)
- 958 repeat buyers (5+ purchases) subscribe

Customer Segmentation

Loyal

3,116 customers

Returning

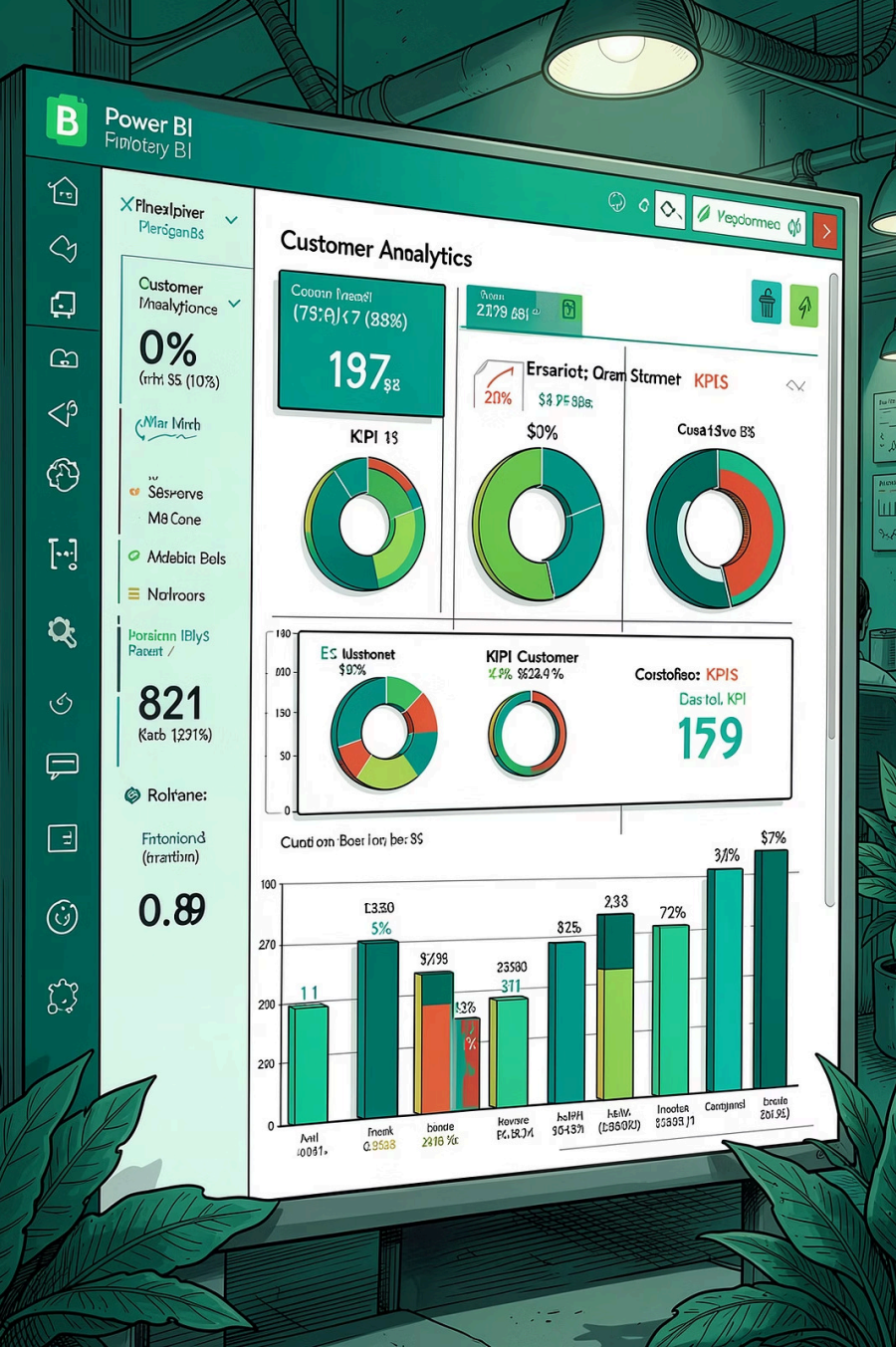
701 customers

New

83 customers

CUSTOMER LOYALTY PROGRAM





Power BI Dashboard

3.9K
Customers

\$59.76
Avg Purchase

3.75
Avg Rating



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward "Loyal" segment



Review Discounts

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers